

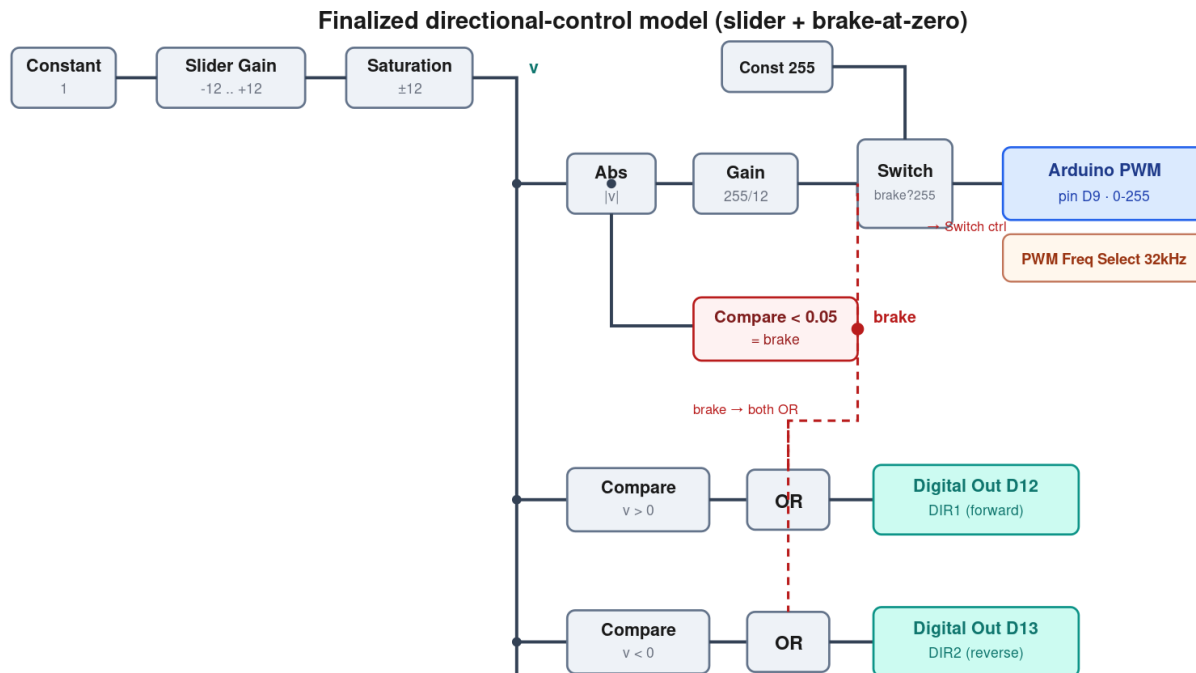
MSE 312 - Directional Control in Simulink (slider gain + brake)

Hand-driven open-loop drive for the TB6568KQ / SN7400 4-gate board (D9 PWM, D12 DIR1, D13 DIR2). Slider sign selects direction, magnitude sets speed, and slider-at-zero commands a short brake. Coast is intentionally not used.

1. Concept

A Slider Gain produces a commanded voltage v from -12 to +12 V. The **sign of v** picks which direction pin is HIGH; the **magnitude $|v|$** maps to the PWM duty. When v is at (or very near) zero, both direction pins are forced HIGH and the PWM is forced to full so the driver sees H,H = short brake. Coast is intentionally not used.

2. Model signal flow



$v > 0$: fwd drive · $v < 0$: rev drive · $v = 0$: both DIR high + PWM 255 = SHORT BRAKE
All blocks + solver fixed-step 0.01 s · External mode (Monitor & Tune)

3. Blocks and parameters

Block (library)	Parameter	Wired to
Constant (Sources)	value = 1	Slider Gain
Slider Gain (Math Ops)	Low -12, High +12	Saturation
Saturation (Discontinuities)	±12	fan-out node v
Abs (Math Ops)	$ v $	Gain + brake compare
Gain (Math Ops)	255/12	Switch input 3
Compare To Constant	operator <, const 0.05	brake flag
Constant (Sources)	value = 255	Switch input 1
Switch (Signal Routing)	criterion $u2 > 0$ (brake)	Arduino PWM
Arduino PWM (Arduino SP)	pin 9	H-bridge PWM

PWM Freq Select (RASPlib)	32 kHz / prescaler 1	(sets D9 freq)
Compare To Constant	operator > 0	OR (DIR1)
Compare To Constant	operator < 0	OR (DIR2)
Logical Operator x2	OR	Digital Output
Digital Output (Arduino SP)	pin 12 / pin 13	DIR1 / DIR2

4. Brake logic (the key detail)

brake = (|v| < 0.05)

D12 = (v > 0) OR brake

D13 = (v < 0) OR brake

PWM = brake ? 255 : (|v| x 255/12)

The Switch forcing PWM to 255 during brake is essential: since IN = PWM AND DIR, if PWM stayed at 0 the outputs would be L,L (coast) even with both DIR pins HIGH. Full PWM makes them H,H (brake).

5. Resulting behavior

Slider v	D12	D13	PWM (D9)	IN1	IN2	Mode
v > 0	1	0	v duty	chop	L	Drive forward
v < 0	0	1	v duty	L	chop	Drive reverse
v ~ 0	1	1	255 full	H	H	Short brake

6. Deploy recipe

Gear -> Model Configuration Parameters -> Hardware Implementation = Arduino Mega 2560. Under Target hardware resources -> Host-board connection, set host COM port Manually to your port. Solver tab: Start 0, Stop inf, Fixed-step, ode5, Fixed step size 0.01 (match every block). Simulation mode = External. Click Monitor and Tune to build, download and run live.

7. First-run safety

- Start with the slider at 0 (brake) before deploying.
- Nudge slightly positive - confirm direction, then negative to confirm the opposite DIR pin takes over.
- Wrong way for the crank-slider? Swap the two motor leads on OUT1/OUT2 - do not rewire the logic.
- Watch supply current; a stalled H,H brake or a short shows up as a current spike.
- Short brake gives zero holding torque at true standstill - a heavy static load can still creep slowly.